

Lab Tasks

SIEM

MI102 - Network Systems and Security

Fachhochschule Kiel

Florian Caspar

Task 7 (1)

- The aim of this task is to learn how to use a SIEM system.
- You may decide on your own which SIEM system you want to use.

Task 7 (2)

- Examples for SIEM Systems:
 - Elastic Stack (trial version, cloud-based)
 - Wazuh (trial version, cloud-based)
 - Wazuh (free, on premise)
 - ELK stack (free, on premise)

Task 7 (3)

- Operate a SIEM system
 - to supervise
- Install three servers
 - to be supervised
 - it can be web servers or any other servers
 - they can be cloned virtual machines
 - you can use the servers from lab task 1

Task 7 (4)

- Configure the SIEM system properly
 - you shall supervise the three servers
 - you shall be alarmed in case of an attack

Task 7 (5)

- Perform an attack on each of the three servers.
- Use a different attack type for each server.
- Demonstrate the attack detection by the SIEM system caused by each attack.
- Show one countermeasure taken after each alarm of the SIEM system.

Task 7 (6)

- Examples for attack types:
 - DDoS Attack
 - Brute Force Attack
 - Change of a file that is supervised with File Integrity Monitoring
 - Creation of root account as backdoor
 - Creation of an SSH server as backdoor
 - Installation of Application with known vulnerabilities

©2023 Florian Caspar